

DETAILED PROJECT DOCUMENTATION

AI-DRIVEN PUBLIC FACILITY UTILIZATION SYSTEM

Review 1 Documentation — Problem Understanding, Research & Initial DSA Design

Field	Details
Course	Data Structures and Algorithms – II
Course Code	CCSE0301
Programme / Branch	B.Tech CSE-A
Assignment Type	Individual Assignment / PBL
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Project	AI-Driven Public Facility Utilization System
SDG	SDG 11 — Sustainable Cities and Communities
Review	Review 1 — Problem Understanding & Initial Progress
Review Coverage	Trees, Graphs, Heaps and Supporting DSA
Current Progress	Analytical and conceptual foundation
Status	Research, problem analysis, DSA applicability study and conceptual design; implementation planned for later review

Note: This document is intentionally more detailed than the 3-page progress report. It is designed to serve as supporting evidence for the work reported in Review 1. The distinction between completed work and proposed/future implementation is maintained throughout.

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1. Introduction

Public facilities such as hospitals, libraries, community centres, parks, sports complexes, public service offices and other shared infrastructure are created to serve a large population. The main difficulty is not only storing facility records. The more important problem is deciding how available facilities should be utilised when demand changes by time, location, capacity, service type and user requirements.

The proposed AI-Driven Public Facility Utilisation System studies this problem from a Data Structures and Algorithms perspective. The system is designed to combine AI-based demand estimation with efficient data organisation and retrieval. Instead of treating the project as only a directory of public facilities, the focus is on the algorithmic layer that can search facilities, model relationships between locations and services, process availability, estimate utilisation, prioritise suitable facilities and return a Top-K recommendation list.

The AI layer can estimate future or current demand from historical utilisation, time, day, location, capacity and other available features. The DSA layer then processes facility records efficiently. Trees can provide indexed retrieval, graphs can represent spatial and service relationships, queues can represent requests and waiting lines, hashing can provide fast ID lookup, and heaps can prioritise the highest-scoring facilities.

The work represented by this Review 1 documentation is primarily analytical and conceptual. It establishes the problem, reviews relevant research, identifies a project-specific gap, studies DSA concepts, selects structures that genuinely fit the problem, and designs the initial AI-plus-DSA workflow. Final model accuracy and measured execution performance should only be claimed after implementation and testing.

The documentation therefore answers five questions:

- Why is public facility utilization a meaningful real-world problem?
- How can AI help predict demand and support better utilisation decisions?
- Why are particular DSA structures useful for facility retrieval, relationships and prioritisation?
- How will the selected DSA concepts work together with the AI component?
- How can the future implementation be tested for correctness, efficiency and usefulness?

2. Project Domain and Context

2.1 Domain

Domain: Intelligent public facility utilisation, demand-aware facility recommendation and data-driven resource allocation.

2.2 Why this domain was selected

Public facilities naturally contain records, relationships, capacity constraints and changing demand. A facility has a type, location, capacity, services, operating hours and historical utilisation. A user has a service requirement and a preferred location or time. Multiple facilities may provide the same service, while one facility may support many services. This creates a practical environment for trees, graphs, queues, heaps, hashing, searching and sorting without adding DSA structures artificially.

Entity	Important attributes	Why it matters
Facility	ID, name, type, location, capacity, services, hours, current occupancy	Main resource being recommended and monitored
Service	Service ID, category, description, eligibility	Connects public needs with facilities
Location	Area, coordinates, zone, nearby facilities	Supports accessibility and graph-based relationships
Utilisation Record	Date, time, occupancy, bookings, service count	Historical evidence for AI demand prediction
User Request	Request ID, service, time, location,	Input used to find a suitable facility

	constraints	
Facility Manager	Facility ID, operating rules, capacity, maintenance status	Uses system insights for planning
Prediction	Forecast demand, confidence, time window	AI output used by the recommendation layer

3. Problem Statement

The proposed system addresses the problem of inefficient and reactive utilisation of public facilities when many facilities and users exist. A simple list of facilities does not answer which facility should be recommended for a specific user request or where under-utilisation and over-crowding are likely to occur.

The problem can be expressed as:

- Given a user request R containing a required service, preferred time, location and constraints.
- Given a collection of public facilities F1...Fn with capacity, availability, service coverage and historical utilisation.
- Given historical and current data from which an AI model can estimate demand.

Determine:

1. Which facilities can satisfy the requested service and hard constraints?
2. What is the predicted or current utilisation level of each suitable facility?
3. Which facility gives the best combination of accessibility, availability, capacity fit, predicted crowding and service suitability?
4. How can the highest-scoring K facilities be retrieved efficiently?
5. How can facility managers identify under-utilised or over-loaded resources?

Output: a ranked Top-K list of suitable public facilities together with an explainable utilisation/recommendation score and relevant demand information.

3.1 Why manual selection becomes difficult

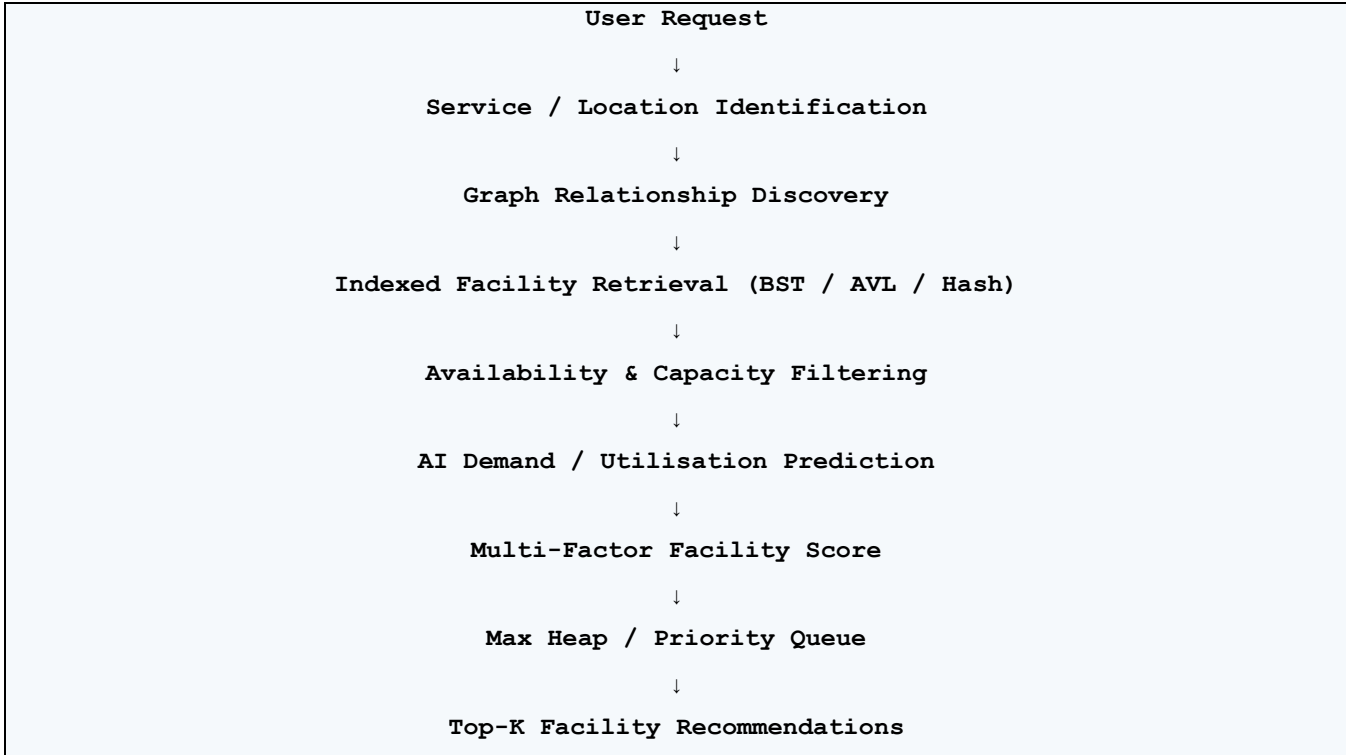
- A user may need to compare many facilities offering similar services.
- Facility availability and occupancy can change by hour or day.
- The nearest facility is not always the best facility if it is already overcrowded.
- Historical utilisation and predicted demand must be considered together.
- Capacity, service type, operating hours and maintenance status create multiple constraints.
- Repeatedly scanning every facility becomes less attractive as the system grows.
- Facilities, locations and services form many-to-many relationships that are difficult to represent with a simple flat list.

4. Problem Understanding and Identification

Area	Understanding
Existing problem	Public facilities may be unevenly utilised: some are overcrowded while others remain under-used.
Users affected	Citizens, visitors, facility managers, municipal administrators and service planners.
Core information	Facility type, service, capacity, location, operating hours, occupancy, historical utilisation and demand prediction.
Core operations	Search, filter, predict, score, rank, allocate and monitor.
Main DSA need	Fast retrieval, relationship modelling, request handling, priority-based recommendation and efficient reporting.
AI role	Estimate demand/utilisation and identify patterns from historical and contextual data.
Research direction	Combine AI demand intelligence with transparent DSA-based

4.1 Problem formulation

The system treats public facility utilisation as a pipeline rather than as one algorithm. First, the user requirement is converted into service and location attributes. Next, related facilities are discovered and indexed records are retrieved. Hard constraints are applied. The AI component estimates demand/utilisation. A multi-factor score is then calculated and a priority structure returns the best facilities.



5. Existing Public Facility Management and Research Background

Existing smart-city and facility-management systems increasingly use sensors, mobile applications, historical usage records, dashboards and predictive analytics. The purpose of this project is not to claim that existing systems lack AI or recommendation features. Instead, the project defines an academic DSA contribution: how facility data and relationships can be organised so that AI predictions can be converted into efficient, explainable facility recommendations.

Aspect	Existing system direction	Project focus
Facility discovery	Search, maps, categories and filters	Efficient indexed retrieval using Hash/BST/AVL
Demand analysis	Dashboards and predictive analytics	AI demand/utilisation score used inside recommendation pipeline
Relationships	Maps and service-area information	Graph of facilities, services, locations and transport/nearby links
Requests	Bookings, appointments and service requests	Queue-based request handling and waiting-list support
Recommendations	Distance, availability or personalised suggestions	Transparent multi-factor score + Top-K priority ranking
Utilisation	Historical reports and current occupancy	Combine historical utilisation with AI prediction and capacity
Academic contribution	Complex commercial/proprietary	Understandable DSA-based

	implementations	implementation suitable for PBL evaluation
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The scope is intentionally limited to a transparent prototype. The project can later be expanded to real-time sensor streams, geospatial models, advanced machine learning, reinforcement learning or a production-scale cloud architecture.

6. Literature Review — Three Selected Papers

6.1 SmartSport: crowd counting meets large language models for smart facility management (2026)

This research presents an intelligent management framework for public sports facilities that combines automated crowd counting with a large language model based analysis module. The system uses usage data, geographic information, demographic context and other information to recognise patterns, diagnose supply-demand gaps and generate management recommendations. The work is directly relevant because it shows the value of moving from manual utilisation observation toward automated monitoring and intelligent decision support.

Relevance to this project: the proposed system similarly treats utilisation as a data-driven decision problem. The difference is that this PBL project concentrates on the DSA layer—efficient retrieval, relationship modelling, request processing and Top-K recommendation—while the AI component supplies demand/utilisation intelligence.

6.2 The satisfaction of ecological environment in sports public services by artificial intelligence and big data (2025)

This paper by Ke Mu, Zhiling Wang, Jinzhou Tang, Jiarui Zhang and Wenxia Han studies the relationship between AI-driven sports public services and residents’ satisfaction with the fitness ecological environment. It combines a CNN model using residual modules and attention mechanisms with the SERVQUAL evaluation model and analyses data from questionnaires, literature reviews and interviews. The study also discusses how AI can monitor sports venues and public fitness facilities, identify overcrowding or equipment problems, and support adjustments to facility use and service planning.

Relevance to this project: the paper directly supports the idea that AI and big-data analysis can improve the quality and utilisation of public facilities and services. Our project extends this direction from sports facilities to a broader public-facility utilisation system by combining predicted utilisation, capacity, availability and accessibility with DSA-based retrieval and Top-K recommendation.

6.3 AI-Driven Optimization of Sanitary Facilities in Office Buildings: A Machine Learning Approach Using LSTM Neural Networks (2025)

This paper by Samson Tan and Teik Toe Teoh presents an AI-based approach for planning sanitary facilities in office buildings using Long Short-Term Memory (LSTM) neural networks. The study examines occupancy patterns, daily fluctuations and building characteristics to provide data-driven guidance for facility planning. It validates the approach using empirical data from a Singaporean office building and compares the LSTM approach with traditional methods. The research demonstrates the value of time-series prediction for facility demand and resource planning.

Relevance to this project: this paper provides a strong example of using AI prediction to convert historical and time-dependent facility usage into planning decisions. Our system generalises the same principle to public facilities: historical utilisation and contextual data can be used to estimate demand, after which DSA structures retrieve suitable facilities, filter constraints and rank the best options.

6.4 Paper-to-project comparison

Paper	Main direction	What this project takes
SmartSport (2026)	Crowd counting + LLM-based intelligent sports facility management	Usage monitoring, demand-gap identification and actionable facility-management recommendations

Sports public-service satisfaction study (2025)	AI/deep learning + big data + SERVQUAL for sports public-service satisfaction	AI-based service-quality evaluation, facility monitoring and understanding user satisfaction
Sanitary facilities with LSTM (2025)	LSTM-based prediction of sanitary-facility demand using occupancy and time-dependent factors	Time-series demand prediction and data-driven facility/resource planning

7. Research Gaps Identified

The gaps below distinguish literature observations from project-specific design decisions. The reviewed literature motivates intelligent facility management and allocation; it does not prescribe the exact DSA structures used here.

Gap / Observation	Basis	Project response
Static facility selection may ignore changing demand	Sports public-service and facility-management research	Use AI demand/utilisation prediction as one input to facility ranking
Current occupancy and predicted demand may need joint consideration	LSTM sanitary-facility planning study	Combine current availability/capacity with predicted demand
Facility allocation involves spatial and service relationships	Project-specific extension from the reviewed facility studies	Use Graph + Adjacency List to represent facility/service/location relationships
Large facility collections require efficient retrieval	Derived from project scalability requirements	Use Hash/BST/AVL indexing for appropriate retrieval operations
Best facility is not always the nearest or cheapest	Multi-factor facility allocation context	Use a transparent weighted score and Top-K priority ranking
AI predictions need an explainable decision layer	Project-specific requirement	Connect AI predictions to transparent FacilityScore and Top-K DSA ranking
DSA and AI are often discussed separately	Project-specific synthesis	Connect AI output to DSA retrieval, filtering and heap-based recommendation

7.1 Core project gap statement

The project explores how AI-based utilisation and demand estimation can be extended into a broader public-facility recommendation pipeline. Unlike the sports-service satisfaction study, which concentrates on sports public services and ecological-environment satisfaction, and the sanitary-facility study, which focuses on office-building restroom planning with LSTM, this project aims to provide a common facility-utilisation framework across different public services. The proposed contribution is the integration of AI prediction with transparent DSA-based retrieval, relationship modelling and Top-K facility recommendation.

7.2 Important viva clarification

The reviewed research papers do not specifically prescribe AVL trees, adjacency lists, hash tables or Max Heaps for this project. Those structures are selected by the project based on required operations. Literature motivates the public-facility utilisation problem and AI direction; DSA analysis determines how the prototype will organise and process the data.

8. Project Objectives

1. Understand the public facility utilisation problem and its practical challenges.

2. Study existing AI-driven facility monitoring, forecasting and allocation approaches.
3. Identify a project-specific research gap around transparent AI + DSA facility recommendation.
4. Represent facility, service, location and utilisation data in suitable structures.
5. Use Hashing and indexed trees for efficient facility lookup where appropriate.
6. Use Graphs and an Adjacency List to model relationships among facilities, services and locations.
7. Use BFS/DFS to discover nearby or related facilities and service connections.
8. Use sorting/searching as baseline and reporting operations.
9. Integrate an AI demand/utilization prediction score with DSA-based retrieval and ranking.

9. Stakeholders and User Requirements

Stakeholder	Need	Expected system support
Citizens / Users	Find a suitable public facility without comparing many options manually	Search, filter, predicted crowd level and Top-K recommendations
Facility Managers	Understand demand and improve utilisation	Predicted utilisation, overload/under-use alerts and facility ranking
Municipal Administrators	Allocate resources across facilities	Area-wise demand, utilisation comparison and planning insights
Service Planners	Identify service gaps and overloaded locations	Graph-based relationship analysis and supply-demand indicators
System Administrator	Maintain facility and service data	Structured records, indexing, logs and performance monitoring

9.1 User-side requirement flow

1. User selects a required public service.
2. User provides preferred location/area and time.
3. System retrieves facilities that offer the requested service.
4. Basic constraints such as operating hours, maintenance and capacity are applied.
5. AI estimates demand/utilisation for candidate facilities.
6. System calculates a multi-factor facility suitability score.
7. Top-K facilities are returned with an explanation.

10. Functional and Data Requirements

10.1 Functional requirements

- Accept facility search and utilisation requests.
- Store public facility, service, location and utilisation records.
- Identify facilities that provide the requested service.
- Retrieve facilities using efficient indexes.
- Apply hard constraints such as operating hours, maintenance and capacity.
- Generate or receive AI demand/utilisation predictions.
- Calculate a multi-factor facility suitability score.
- Rank facilities and return Top-K recommendations.
- Provide explainable score components.
- Maintain request/waiting queues where required.
- Generate utilisation reports for facility managers.
- Measure algorithmic and system performance during later implementation.

10.2 Proposed facility record

```
Facility {  
    id  
    name  
    type  
    location  
    services[]  
    capacity  
    operatingHours  
    currentOccupancy  
    maintenanceStatus  
    historicalUtilisation[]  
}
```

10.3 Proposed request record

```
FacilityRequest {  
    requestId  
    serviceRequired  
    preferredLocation  
    requestedTime  
    expectedUsers  
    constraints[]  
}
```

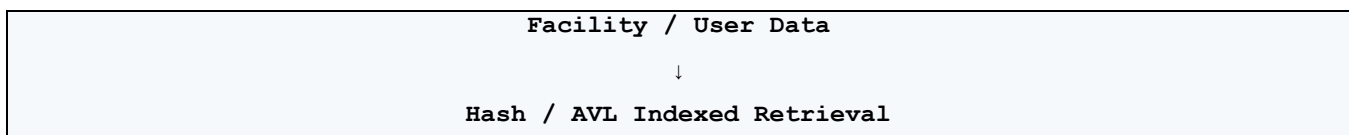
10.4 Why these attributes were selected

Attribute	Purpose in utilisation/recommendation
Service	Checks whether the facility can satisfy the user's need
Location	Supports accessibility and graph-based nearby-facility discovery
Capacity	Prevents recommendations that cannot safely handle expected demand
Current Occupancy	Represents the present load on a facility
Historical Utilisation	Provides data for AI demand estimation
Operating Hours	Removes facilities unavailable at the requested time
Maintenance Status	Prevents unavailable facilities from being recommended
Predicted Demand	Estimates future/current load and possible overcrowding
Accessibility / Distance	Represents practical suitability

11. DSA-II Concepts Studied, Selection, Implementation and Rejection Analysis

The project does not include every DSA concept merely to satisfy a syllabus checklist. Each concept is mapped to a concrete operation: storing data, fast lookup, ordered search, relationship modelling, request processing, traversal, ranking, reporting or utilisation analysis. Some structures are core, some are supporting, and some are deliberately not selected.

Overall proposed pipeline:





11.1 DSA selection philosophy

The system requires several different algorithmic capabilities. First, facility records must be retrieved efficiently. Second, service and location relationships must be represented because one service may be available at many facilities and facilities may be connected through areas or nearby links. Third, requests may need FIFO processing. Fourth, after scoring candidate facilities, the highest-scoring facilities must be retrieved efficiently. These needs motivate hashing, trees, graphs, queues and heaps.

DSA concept	Role	Decision
Binary Tree	Foundation for tree concepts	Studied, not core
BST	Baseline ordered facility search	Baseline
AVL Tree	Balanced indexed retrieval	CORE planned
Threaded Binary Tree	Stack-free traversal	Not core
Min Heap	Lowest occupancy/cost/priority queries	Optional
Max Heap	Highest suitability/utilisation priority	CORE
Graph	Facility-service-location relationships	CORE
Adjacency List	Sparse graph representation	CORE
Adjacency Matrix	Dense graph comparison	Alternative
BFS / DFS	Graph exploration	CORE supporting
Searching	Baseline lookup and comparisons	Supporting
Sorting	Reports, baseline ranking and analysis	Supporting

12. Detailed Tree-Based Analysis

12.1 Why a tree is considered

A public-facility system may contain thousands of facility records. A tree can maintain an ordered index over identifiers, names, zones or other selected keys. The choice of tree matters because a normal binary tree does not guarantee an ordered search path.

12.2 Binary Tree — why it is not the primary structure

A binary tree permits at most two children per node but does not inherently define an ordering relationship. Searching for a facility by ID may therefore require visiting many nodes and can take $O(n)$ in the worst case. It is academically important because it is the foundation for BST and AVL structures, but it is not the main indexed-search structure of this project.

12.3 Binary Search Tree

A Binary Search Tree adds the ordering rule that smaller keys are placed on the left and larger keys on the right. This allows directed search. In an approximately balanced BST, search, insertion and deletion are $O(\log n)$ on average, while a skewed tree can degrade to $O(n)$.

```
searchFacility(root, key):
    if root == NULL: return NOT_FOUND
    if key == root.key: return root
    if key < root.key: return searchFacility(root.left, key)
    else: return searchFacility(root.right, key)
```

Example: if facility IDs F103, F101, F105, F100 and F102 are inserted, F103 becomes the root. A search for F102 can follow $F103 \rightarrow F101 \rightarrow F102$ instead of scanning every record.

12.4 AVL Tree — preferred balanced tree

An AVL Tree is a self-balancing BST. The balance factor is $\text{height}(\text{left subtree}) - \text{height}(\text{right subtree})$, and each node is maintained at -1 , 0 or $+1$ after rebalancing. This prevents the long-chain behaviour of a poorly ordered BST and gives guaranteed $O(\log n)$ search, insertion and deletion.

AVL operation	Purpose in project	Complexity
Search	Find facility using an indexed key	$O(\log n)$ guaranteed
Insert	Add new facility record	$O(\log n)$ guaranteed
Delete	Remove inactive facility	$O(\log n)$ guaranteed
Inorder traversal	Produce facilities in sorted key order	$O(n)$

Rotation cases: LL imbalance requires a right rotation; RR requires a left rotation; LR requires a left rotation on the child followed by a right rotation; RL requires a right rotation on the child followed by a left rotation.

12.5 Threaded Binary Tree

A Threaded Binary Tree uses selected null child pointers to store links to inorder predecessor or successor nodes. It can make repeated inorder traversal possible with reduced dependence on recursion or an explicit stack. The present system does not have extremely frequent stack-free inorder traversal as a dominant requirement, so it is studied but not selected as a core structure.

12.6 Tree traversal

Traversal	Complexity	Possible role
Inorder	$O(n)$	Sorted facility output from BST/AVL
Preorder	$O(n)$	Structural inspection or export
Postorder	$O(n)$	Bottom-up processing / cleanup
Level-order	$O(n)$	Breadth-wise tree inspection using a queue

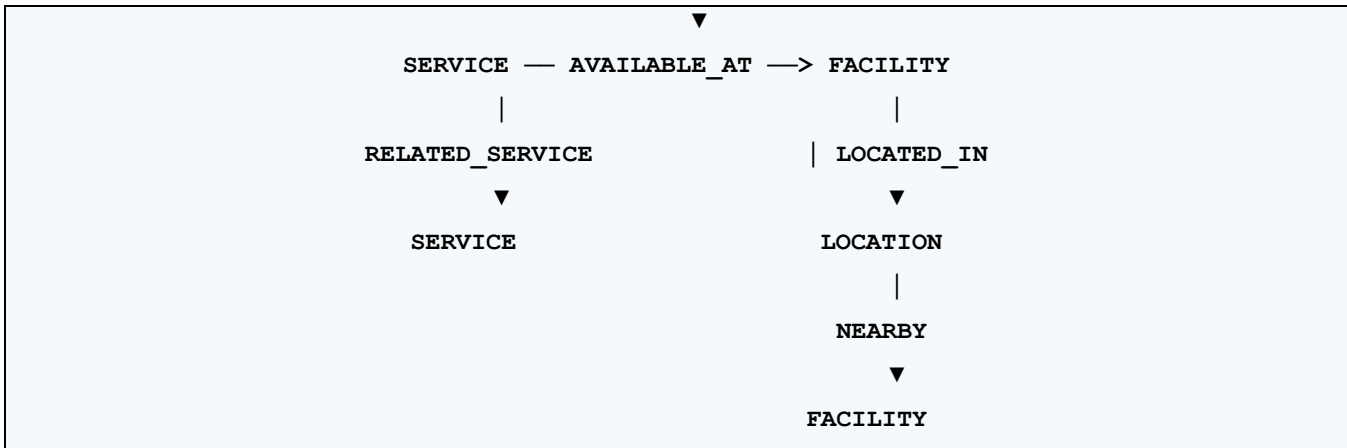
13. Detailed Graph-Based Analysis

13.1 Why a graph is necessary

The system contains many-to-many relationships. One facility may offer many services, one service may be available at many facilities, facilities may be connected by nearby locations, and facilities may belong to service zones. A graph represents these cross-connections naturally, while a tree is constrained by hierarchical parent-child relationships.

13.2 Proposed graph model

USER REQUEST
REQUIRES



13.3 Graph components

Component	Representation
Vertex	Facility, Service, Location or Request
Edge	AVAILABLE_AT, REQUIRES, LOCATED_IN, NEARBY, RELATED_TO
Weight (optional)	Distance, travel time, relevance, capacity gap or similarity
Adjacency List	For each vertex, store directly connected vertices/edges

13.4 Adjacency List vs Adjacency Matrix

Aspect	Adjacency List	Adjacency Matrix
Space	$O(V+E)$	$O(V^2)$
Best suited for	Sparse graphs	Dense graphs
Project fit	Preferred because each facility connects to a subset of all possible entities	Less preferred for a large sparse facility graph
Edge check	$O(\text{degree})$	$O(1)$
Traversal	$O(V+E)$	$O(V^2)$

Max Heap

A Min Heap is useful when the system wants the smallest value, such as the lowest predicted occupancy, lowest distance or lowest waiting time. It is optional because the main recommendation objective is not simply the minimum of one attribute. A Max Heap is the core priority structure because the system needs repeated access to the highest facility suitability score.

Example: if facilities have suitability scores F101=91, F102=74, F103=88, F104=95 and F105=81, a Max Heap returns F104, F101 and F103 for Top-3.

15. Proposed AI-Driven Utilization Method

15.1 Why AI is required

DSA structures organise and process data efficiently, but they do not predict future demand. The AI component is therefore responsible for estimating expected facility utilisation from historical and contextual information. The DSA component then uses that prediction as one factor in the recommendation and allocation pipeline.

15.2 Proposed AI input features

- Historical occupancy or usage count.
- Hour of day and day of week.

- Season or special-event indicator where available.
- Facility capacity and service type.
- Recent utilisation trend.
- Location or zone demand indicator.
- Current occupancy and availability.
- Optional weather, holiday or event features if reliable data are available.

15.3 AI output

The AI model can output a predicted utilisation percentage or expected number of users for a selected facility and time window. The output can be converted into a demand state such as Low, Moderate, High or Overloaded. The exact model and weights should be finalised experimentally rather than assumed in Review 1.

15.4 Proposed facility suitability score

A transparent weighted score is proposed so that AI predictions remain understandable to users and evaluators.

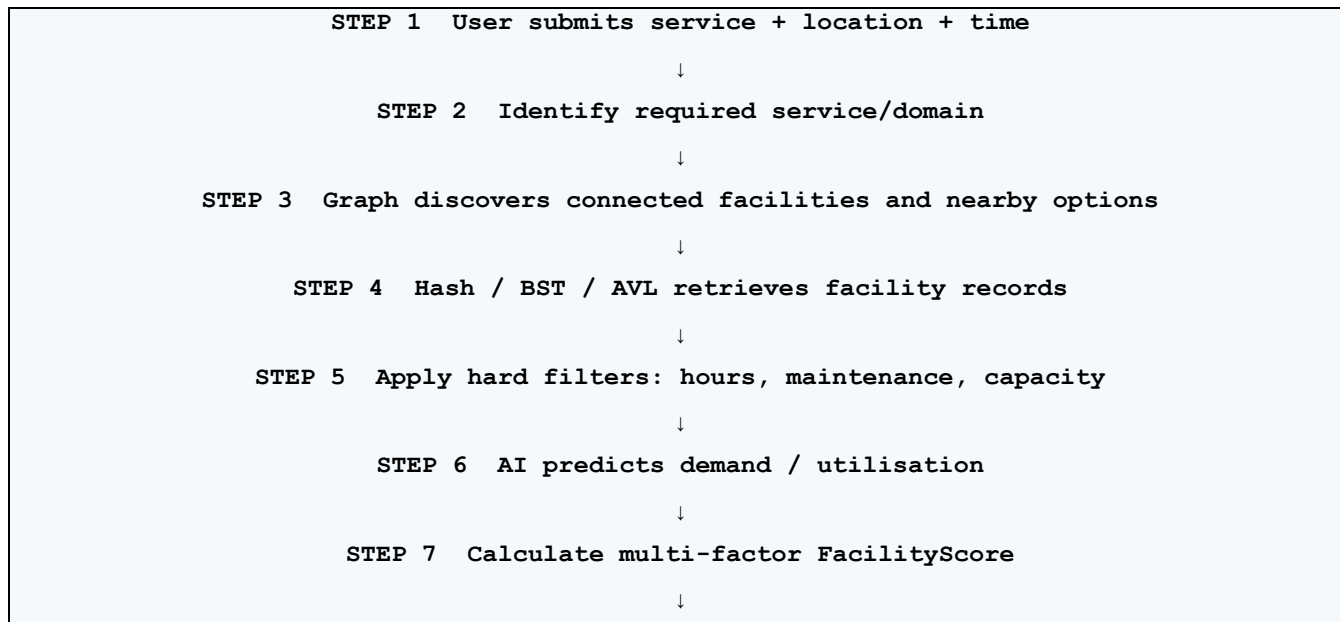
```
FacilityScore =
    w1 * ServiceMatch
  + w2 * AvailabilityScore
  + w3 * CapacityFit
  + w4 * AccessibilityScore
  + w5 * DemandSuitability
  + w6 * HistoricalUtilisationScore
  + w7 * FacilityQualityScore
```

DemandSuitability should reward a facility that can serve the request without being predicted to become overloaded. For example, a facility with high service compatibility but extremely high predicted occupancy may receive a lower final score than a slightly farther facility with sufficient capacity.

15.5 Explainability

The system should show why a facility was recommended. An explanation can include service match, predicted demand level, available capacity, distance/accessibility and operating status. This is important because a public-service recommendation should be understandable rather than a black-box number.

16. Proposed End-to-End System Workflow



STEP 8 Insert candidates into Max Heap

↓

STEP 9 Extract highest-scoring K facilities

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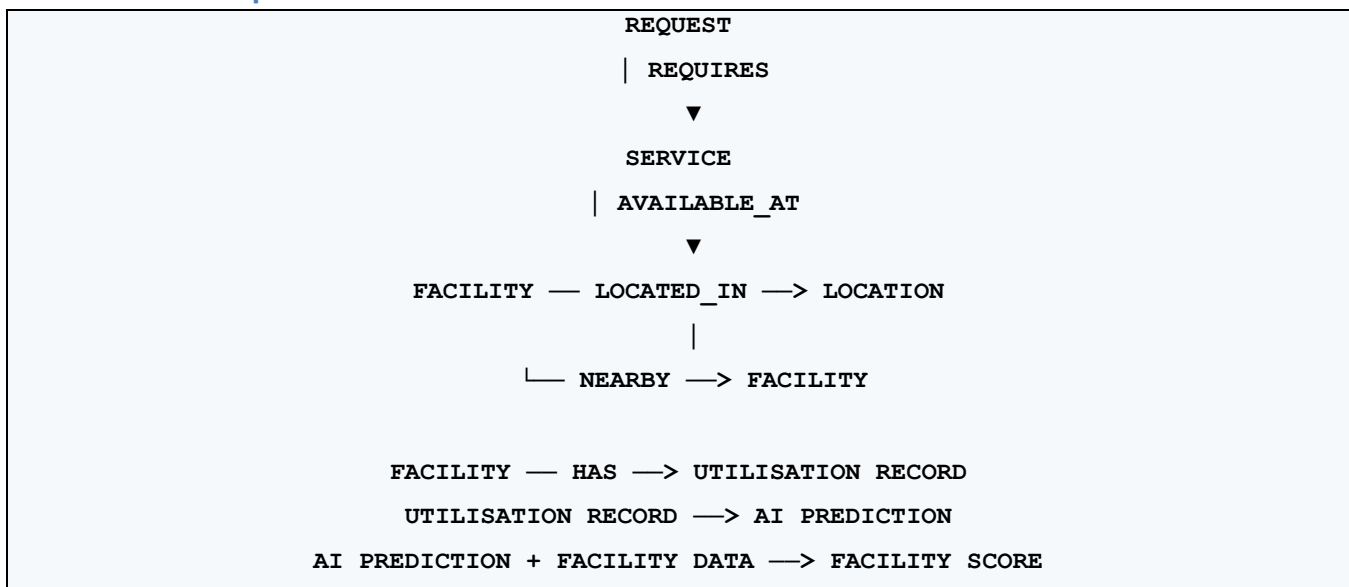
STEP 10 Return recommendation + explanation + utilisation status

16.1 Why the stages are separated

Stage	Reason for separation
Requirement analysis	Converts user needs into usable service/location attributes
Graph exploration	Captures service and spatial relationships
Candidate retrieval	Avoids scoring clearly irrelevant records
Filtering	Removes facilities that cannot serve the request
AI prediction	Estimates demand/utilisation
Scoring	Produces a comparable numerical suitability value
Heap ranking	Efficiently prioritises high-scoring facilities
Explanation	Makes the decision understandable
Evaluation	Separates correctness and usefulness from execution speed

17. Conceptual Data Model

17.1 Relationship model



17.2 Example record set

Facility	Service	Capacity	Current occupancy	Predicted demand	Status
F101 Community Library	Library / Study	120	68	75%	Available
F102 Central Sports Hall	Indoor Sports	200	176	94%	High Load
F103 Community Centre	Community Services	150	52	48%	Available

For a request requiring a public study facility, F101 may be preferred over a more crowded facility even if both provide related services. The graph identifies candidates, indexed structures retrieve their records, AI estimates demand, and the scoring/heap stage produces the final ranking.

18. Proposed Algorithms and Pseudocode

18.1 Facility retrieval using BST/AVL

```
searchFacility(root, key):
    if root == NULL:
        return NOT_FOUND
    if key == root.key:
        return root
    if key < root.key:
        return searchFacility(root.left, key)
    else:
        return searchFacility(root.right, key)
```

18.2 Graph traversal using BFS

```
BFS(start):
    create queue Q
    mark start visited
    push start into Q
    while Q is not empty:
        u = pop Q
        process u
        for each v in adjacencyList[u]:
            if v is not visited:
                mark v visited
                push v into Q
```

18.3 Demand-aware facility score

```
calculateScore(request, facility, prediction):
    service = serviceMatch(request, facility)
    availability = availabilityMatch(request, facility)
    capacity = capacityFit(request, facility)
    access = accessibilityScore(request, facility)
    demand = demandSuitability(prediction, facility)
    history = historicalUtilisationScore(facility)
    quality = facilityQualityScore(facility)
    return w1*service + w2*availability + w3*capacity
        + w4*access + w5*demand + w6*history + w7*quality
```

18.4 Top-K using Max Heap

```
for each candidate facility:
    score = calculateScore(request, facility, prediction)
    maxHeap.push(score, facility)
repeat K times:
    result.append(maxHeap.extractMax())
return result
```

18.5 Queue-based request handling

```
submitRequest(request) :  
    requestQueue.enqueue(request)  
  
processRequests() :  
    while requestQueue is not empty:  
        request = requestQueue.dequeue()  
        recommendation = generateRecommendation(request)  
        storeResult(recommendation)
```

19. Complexity and Performance Considerations

Component	Typical complexity	Why it matters
BST search	$O(\log n)$ average; $O(n)$ worst	Baseline ordered search
AVL search/insert/delete	$O(\log n)$ guaranteed	Predictable performance as facilities grow
Tree traversal	$O(n)$	Every relevant indexed node may be processed
Queue enqueue/dequeue	$O(1)$ with suitable implementation	Request and BFS processing
BFS / DFS	$O(V+E)$	Efficient exploration of sparse facility relationships
Adjacency List space	$O(V+E)$	Stores actual relationships rather than all possible pairs
Adjacency Matrix space	$O(V^2)$	Can become expensive for large sparse graphs
Heap peek	$O(1)$	Immediate access to best candidate
Heap insert	$O(\log n)$	Maintains priority ordering
Heap extractMax	$O(\log n)$	Retrieves next best candidate
Full sorting	$O(n \log n)$ typical	Useful baseline/reporting operation

19.1 End-to-end performance view

The final recommendation time depends on candidate discovery, index retrieval, AI prediction time and ranking. The DSA design aims to reduce unnecessary scanning before the scoring stage. For example, graph traversal can identify relevant facilities, hashing or AVL can retrieve records, filtering can remove unsuitable facilities, and a heap can avoid repeatedly sorting the complete candidate set when only Top-K output is required.

19.2 Evaluation planned for later reviews

- Compare linear search, BST and AVL lookup time as the number of facilities increases.
- Measure hash lookup performance for exact IDs.
- Measure BFS/DFS time as vertices and edges increase.
- Compare full sorting against Max Heap for Top-K retrieval.
- Measure recommendation generation time for different candidate-set sizes.
- Measure AI prediction error using appropriate metrics after a dataset is available.
- Check Top-K ranking correctness against a manually verified test set.
- Measure overload/under-utilisation detection quality where labelled data are available.

20. Current Progress

The current stage is the analytical and conceptual foundation. The purpose is to establish a technically justified design before implementation. No final accuracy, speed or utilisation-improvement percentage is claimed without implementation and evaluation evidence.

Work area	Status
Problem understanding	Completed conceptually
Stakeholder and requirement analysis	Completed conceptually
Existing public-facility research study	Completed at literature level
Research-gap identification	Completed
DSA applicability analysis	Completed
Facility-service-location graph model	Designed conceptually
AI demand/utilisation role	Defined conceptually
Multi-factor scoring model	Designed conceptually
Pseudocode	Prepared
Complexity analysis	Prepared
Working application	Planned for later review
Measured AI accuracy	Not yet available
Measured DSA performance	Planned for later review

21. Evidence and Supporting Material

The following items can be used as evidence for Review 1 and can be expanded as implementation progresses.

Evidence item	What it demonstrates
Problem statement and context	Understanding of public facility utilisation and scale-related difficulty
Literature review	Research activity and basis for the proposed AI-driven direction
Research-gap table	Connection between literature observations and project scope
DSA applicability table	Reasoned selection/rejection of syllabus concepts
Facility-service-location graph model	Understanding of many-to-many relationships
AI + DSA workflow	Conceptual integration of prediction, retrieval and ranking
Pseudocode	Initial understanding of how selected operations will be implemented
Complexity table	Awareness of computational cost and scalability
Review 2 plan	Transition from analysis to coding and testing

21.1 Evidence placement

For submission, a concise progress report can remain the official summary while this detailed documentation serves as supporting evidence. During implementation, screenshots of source code, console output, database records, sample predictions, graphs, test results and performance measurements should be added.

22. Challenges Faced and How They Were Addressed

Challenge	Why it was difficult	How it was addressed
Relating DSA to a real problem	It was easy to describe public facilities but harder to identify meaningful algorithmic operations.	Mapped each DSA concept to search, relationship modelling, request handling, prediction support or ranking.
Avoiding forced DSA usage	Not every syllabus structure provides equal value for this problem.	Classified structures as core, supporting, optional or not preferred.
BST vs AVL	Both provide ordered search and may appear redundant.	Used BST as the baseline and AVL as the balanced implementation for predictable performance.
Tree vs Graph	Both organise data but solve different structural problems.	Used trees for ordered/indexed records and graphs for many-to-many service/location relationships.
Min Heap vs Max Heap	Both are priority structures but prioritise	Selected Max Heap for highest

	opposite directions.	suitability; Min Heap remains optional for lowest-distance/lowest-load queries.
AI vs DSA responsibility	AI prediction and DSA processing can be confused as the same layer.	Defined AI as the demand/utilisation estimation layer and DSA as the data organisation/retrieval/ranking layer.
Explainability	A prediction alone does not explain why a facility was recommended.	Designed a transparent multi-factor score with visible components.
Avoiding overcomplication	Adding every possible AI or smart-city feature can make the PBL unrealistic.	Kept the first version focused on demand prediction, DSA retrieval, graph discovery and Top-K recommendation.

23. Individual Contribution

As an Individual Assignment, all work documented in Review 1 is treated as individual project work. The contribution includes both research and technical analysis.

- Defined the AI-Driven Public Facility Utilisation System problem statement and scope.
- Studied public facility utilisation, demand imbalance and recommendation challenges.
- Reviewed relevant research on intelligent facility monitoring, forecasting and allocation.
- Identified research observations and formulated the project-specific gap.
- Studied DSA concepts and compared their suitability for the project.
- Designed the facility-service-location graph model.
- Designed the AI demand/utilisation integration with the DSA pipeline.
- Designed the multi-factor facility scoring and Top-K recommendation approach.
- Prepared initial pseudocode, complexity analysis and implementation plan.
- Prepared this detailed Review 1 documentation.

24. Current Status and Progress

The current progress represents completion of the analytical foundation rather than a percentage of a fully implemented software product.

Work area	Status	Evidence in this document
Problem understanding	Completed	Sections 1–4
Stakeholder/requirements analysis	Completed	Sections 9–10
Research background	Completed	Sections 5–7
AI utilisation concept	Completed conceptually	Section 15
Trees study	Completed conceptually	Sections 11–12
Graphs study	Completed conceptually	Sections 11 and 13
Other DSA mapping	Completed conceptually	Section 14
AI + DSA workflow	Designed conceptually	Sections 15–18
Implementation	Not yet completed	Planned for Review 2
Testing/performance evaluation	Not yet completed	Planned for later reviews

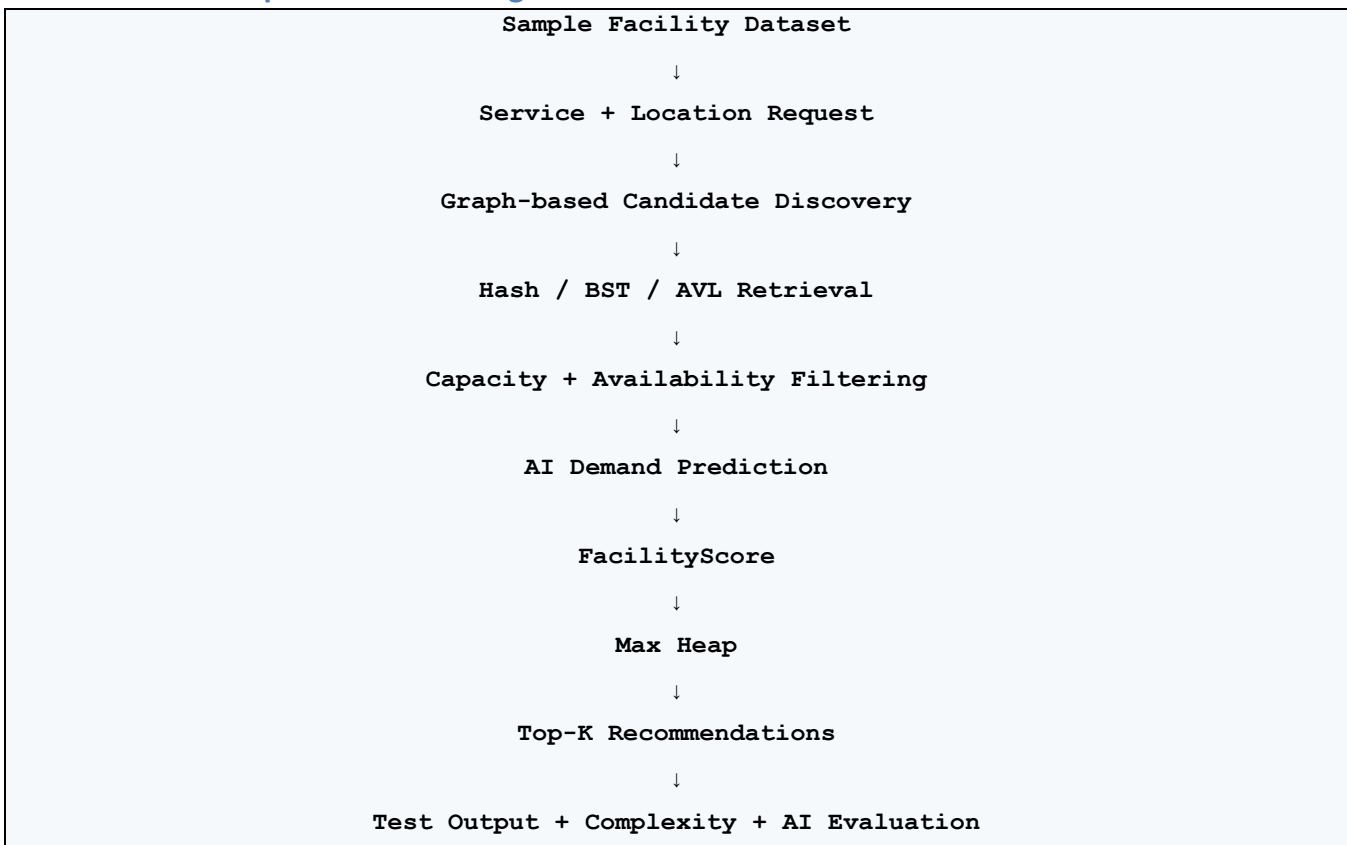
25. Plan for Review 2

The next review should move from conceptual analysis toward a small working implementation. The goal is to implement the selected DSA structures, connect them to a sample public-facility dataset, add the first AI demand/utilisation model and collect measurable evidence.

1. Study the relevant DSA-II topics required for implementation, including graph and heap operations.

2. Finalise the data representation for facilities, services, locations and utilisation records.
3. Create a sample dataset with realistic facility attributes and historical utilisation.
4. Implement Hash-based exact lookup.
5. Implement a small BST baseline and AVL tree for indexed retrieval.
6. Implement the facility-service-location graph using an adjacency list.
7. Implement BFS and DFS tests for candidate discovery and relationship analysis.
8. Implement queue-based request handling.
9. Implement the first version of AI demand/utilisation prediction using a suitable sample dataset.
10. Implement the multi-factor FacilityScore and test different weight configurations.
11. Use a Max Heap to generate initial Top-K recommendations.
12. Add explanations showing the main factors behind each recommendation.
13. Test insertion, search, traversal, filtering, prediction, scoring and ranking operations.
14. Collect evidence: source-code screenshots, diagrams, sample inputs/outputs and timing observations.
15. Compare expected and observed behaviour and document Review 2 outcomes.

25.1 Review 2 implementation target



26. Conclusion

Review 1 establishes the conceptual and algorithmic foundation of the AI-Driven Public Facility Utilisation System. The project begins with the observation that public facilities may be unevenly utilised because demand changes over time and across locations. A useful system therefore needs more than a static facility directory; it should combine current availability, capacity, historical usage and predicted demand.

The literature reviewed for this documentation provides complementary perspectives. SmartSport demonstrates intelligent monitoring and analysis for public sports facilities; the sports-public-service satisfaction study connects AI, big data, public sports services and residents' satisfaction; and the LSTM sanitary-facility study demonstrates time-

series prediction for facility planning and resource optimisation. Together, these papers support the project's AI-driven public-facility utilisation direction.

From the DSA perspective, the analysis concludes that Hashing is useful for direct ID lookup; BST provides a baseline ordered-search structure; AVL provides balanced indexed retrieval; Graphs and Adjacency Lists represent many-to-many service and location relationships; BFS and DFS support graph exploration; Queues support request and waiting-list processing; sorting and searching provide baseline/reporting operations; and Max Heap provides a natural structure for Top-K facility recommendation. Min Heap, Threaded Binary Tree, Adjacency Matrix and some other concepts remain optional or non-core where their main advantages do not match the project's dominant operations.

The key design principle is separation of responsibilities. AI predicts demand or utilisation. DSA structures organise, retrieve, traverse and prioritise the data. The scoring layer combines these signals into an explainable facility suitability score. The final heap-based stage produces the highest-ranked facilities. This separation makes the system easier to implement, test and explain in a DSA course.

The project is intentionally not presented as fully implemented at this stage. The next step is to convert the conceptual model into a small working prototype, validate the selected DSA operations, connect an AI demand model to sample facility data, generate Top-K recommendations and collect measurable evidence for later reviews.

27. References

- Li, P., & Xu, C. (2026). SmartSport: crowd counting meets large language models for smart facility management. *Scientific Reports*, 16, Article 13991. DOI: 10.1038/s41598-026-44145-9.
- Mu, K., Wang, Z., Tang, J., Zhang, J., & Han, W. (2025). The satisfaction of ecological environment in sports public services by artificial intelligence and big data. *Scientific Reports*, 15, 12748. DOI: 10.1038/s41598-025-97927-y.
- Tan, S., & Teoh, T. T. (2025). AI-Driven Optimization of Sanitary Facilities in Office Buildings: A Machine Learning Approach Using LSTM Neural Networks. *Applied Sciences*, 15(5), 2499. DOI: 10.3390/app15052499.
- Babu, H., Al-Shuwaiki, N. M., & Sophika, S. (2025). Real-Time Monitoring and Resource Allocation in Public Administration Using AI-Driven Demand Forecasting: A Comprehensive Review. *Proceedings of International Conference on Applied Innovation in IT*, 13(4), 349–357. DOI: 10.25673/122138.
- Research direction and DSA structure selection in this document are project design decisions based on the operations required by the AI-Driven Public Facility Utilisation System; the reviewed papers are used to motivate the domain and research direction rather than to claim that they prescribe AVL trees, graphs or heaps.

Source note: The supplied reference document was used for the overall documentation structure, depth, DSA mapping style, distinction between completed analysis and future implementation, pseudocode format, complexity discussion and Review 2 planning. The project domain and examples in this document have been rewritten for public facility utilisation. The reference document itself describes a different project domain and explicitly separates research motivation from DSA design decisions. □filecite□turn0file0□L320-L340□

